E3: a Semaev oracle that uses the base's algebra does not change the picture

Experiment E3 of `PLAN.md`, run under the protocol frozen in `E3-PROTOCOL.md` (commit `ede431274`) before any measured run. No security claim. Nothing here concerns P-192 or P-224 beyond PLAN §7′s fixed wording. The curves are a restricted prime family (`2^t | p − 1`).

Question and answer

PR #2297 found that, with the MITM oracle, a random-scalar known-log base (`known-log:scalars=random`: multiples of `G`, logarithms free) matches the smallest-abscissa base's yield. The reason is that MITM never uses the abscissa structure. E3 asks whether an oracle that does use the base's algebraic description changes that comparison.

The oracle is new in this PR: `pkm-semaev`, in `ic.pipeline`. For every target it writes Semaev's $S_4(x_1, x_2, x_3, x_R) = 0$ with each `x_i` on the `prime-kummer` base's tower `(x/g)^{2^t} = 1` (Petit–Kosters–Messeng), and F4 over the tower quotient ring decides it. The algebraic base is `prime-kummer`, because `prime-abscissa`'s bound has no low-degree `F_p` description. The answer, on three 24-bit prime-order curves for `t = 2…5` (B = 1 to 19):

1. **The algebra buys no yield.** At equal B the Kummer base and the random-scalar known-log base have identical exact counts N_3 of points that are a sum of three base points, at all 12 (curve, t) cells. The measured hit rates cover N_3 / r in every cell.
2. **What the algebra costs is F4′s linear algebra.** It is unpriced native work, so §5a prices it in the band. At w = 0 the Semaev arm costs less than known-log MITM (S ratio 0.30–0.72, point estimates). At w = 0.1 and w = 1 it costs $6.5 \cdot 10^3$ to $1.8 \cdot 10^{11}$ times more. The order flips inside the band at every t, so under AGENTS.md §5a **the two are unresolved, and there is no ranking for either side.**
3. **No crossover in B.** The break-even weight w^* at which the arms tie falls 49× to 880× per tower level: about 10^{-6} at t = 2, 10^{-8} at t = 3, 10^{-10} at t = 4 and 10^{-13} at t = 5. That is because F4′s cost per target grows 67–406× per level while MITM's grows 2×. No B^* exists in the band at $w \geq 0.1$, and the trend points away from one.
4. **F4 is far above the trivial algorithm on the same system.** The V^3 grid reference decides every one of these systems with $1.6 \cdot 10^3$ (t = 2) to $6.2 \cdot 10^7$ (t = 5) times fewer operations than F4, with identical verdicts.

So for E3′s question: replacing MITM by an oracle that uses the base's polynomial description does not make a structured base worth more than a random known-log set of the same size. It adds a cost that §5a prices and that grows much faster than the table's.

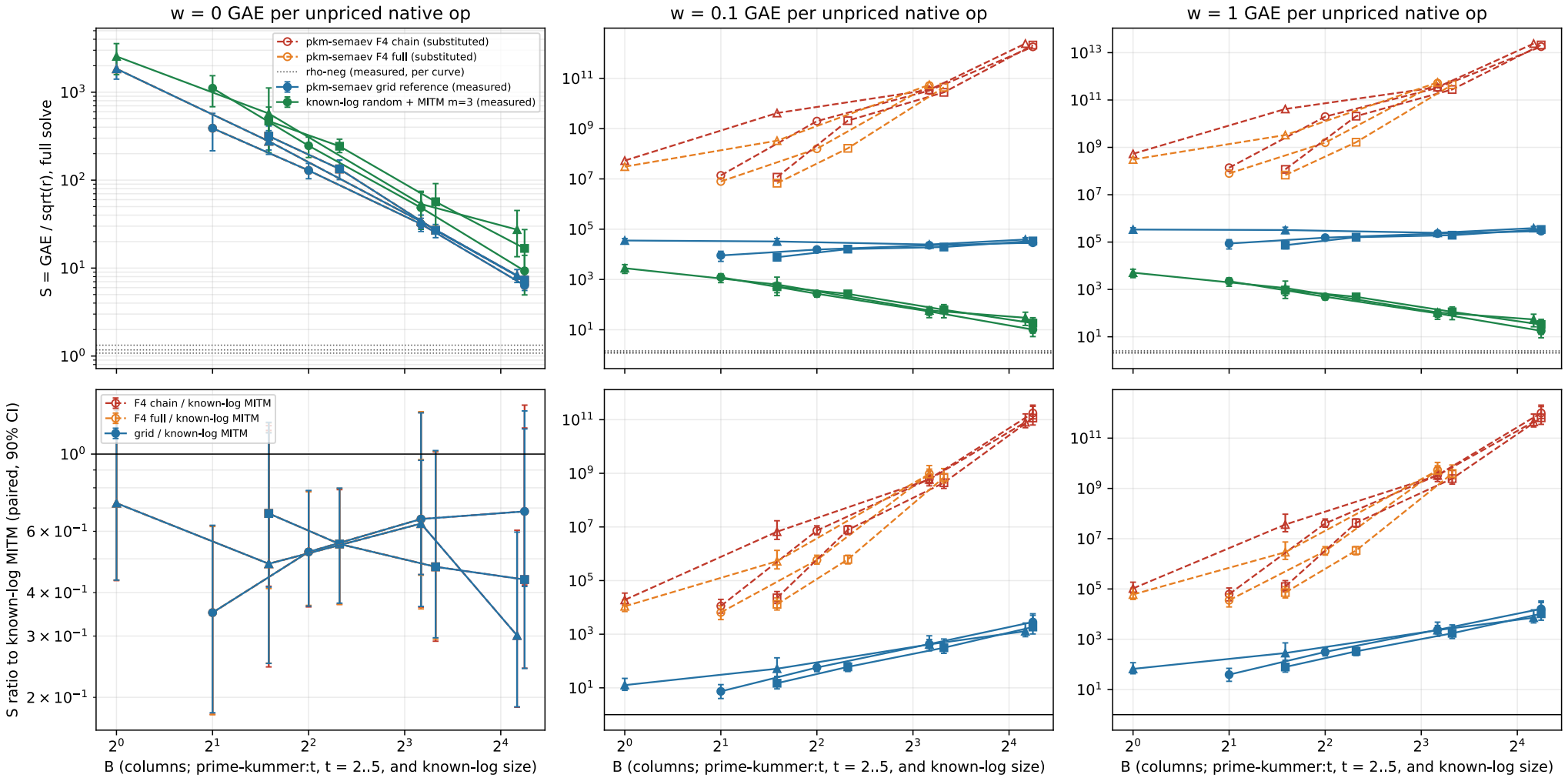


Figure 1: E3: S against B at each band weight (top) and the ratio to known-log MITM (bottom)

Figure 1. S = GAE/√r of a full solve against B, at w = 0, 0.1 and 1 GAE per unpriced native operation. Markers o/s/^ are e3-c1/c2/c3. Solid lines are measured: the grid reference, known-log MITM, and rho-neg (dotted, per curve). Dashed lines are the F4 arms, ***substituted, not run*** *(see Method). At w = 0 the F4 lines coincide with the grid line, because they have the same group work. Error bars are 90% bootstrap intervals over 8 targets, paired by target in the ratios. Sessions ECBS1h642a1a44f9f9, ECBS1hd4736f2baf84, ECBS1h2a56f971e518.*

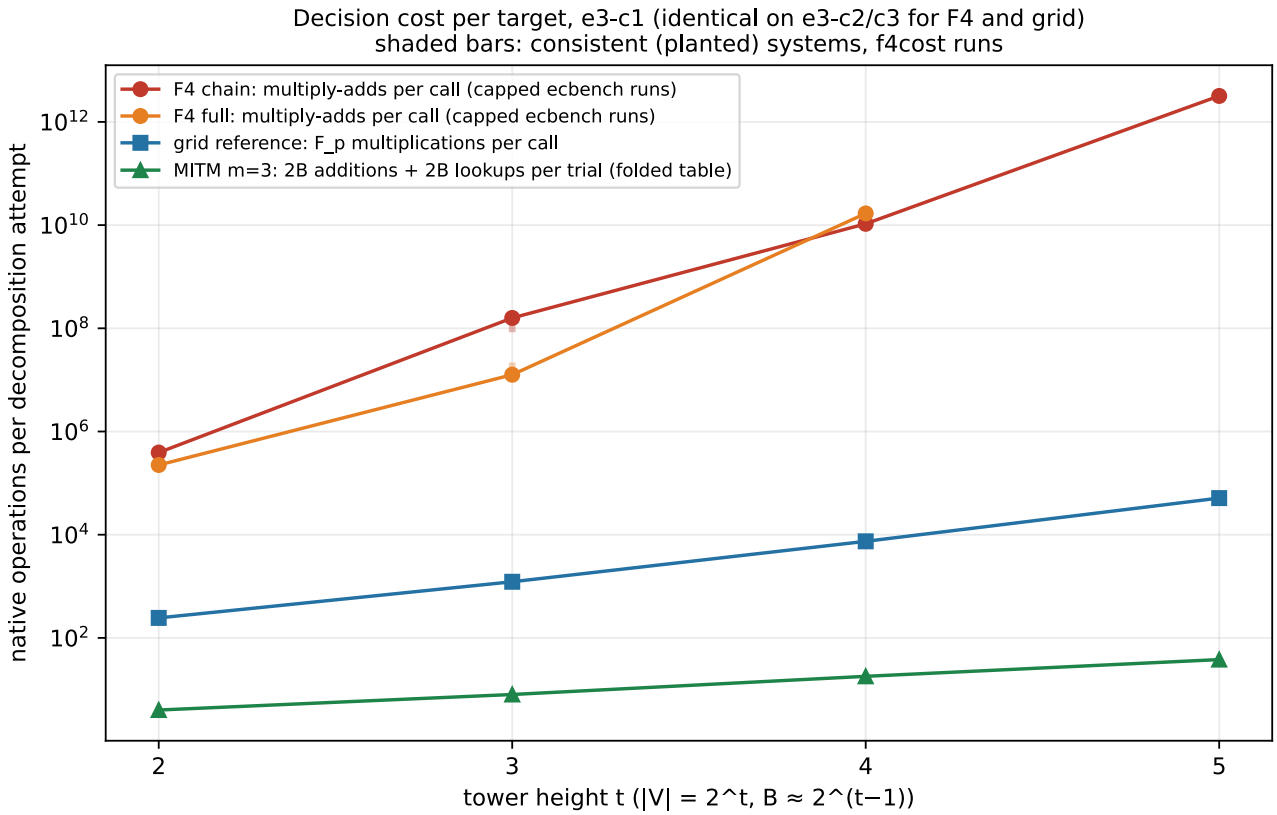


Figure 2: F4 multiply-adds per call against t

Figure 2. Native operations per decomposition attempt against t, on e3-c1. F4′s count per refutation is identical on all three curves at each t. Shaded bars show the planted (consistent) systems from the `f4cost` *runs.*

What ran

item	value
oracle	`pkm-semaev` (`src/cryptanalysis/ic_framework/pkm_semaev.rs`), registered in `solve_ic_prime` (only on `prime-kummer`, whose `t` and `g` it reads) and in `ecbench::claim`
curves	e3-c1 `p = 9061313`, e3-c2 `p = 13040257`, e3-c3 `p = 11223809`: 24-bit, `a = −3`, prime order, $v_2(p-1) \geq 6$ (table in the protocol); control `p = 2393` (12-bit)
arms	per t: `sem-chain` and `sem-full` (F4, capped at `K_t = 400/100/10/1` trials, `crosscheck=1`), `sem-grid` (grid reference, full solve), `kl-mitm3` (`known-log:size=B,scalars=random` + `mitm:m=3,negation_folded=1`, full solve), and `rho-neg`
targets	8 `planted_shared` per curve, target seed 20261019, 1 round, no warm-up, host L0 (shared, no pinning); operation counts only
sessions	e3-c1 `ECBS1h642a1a44f9f9`, e3-c2 `ECBS1hd4736f2baf84`, e3-c3 `ECBS1h2a56f971e518`, control `ECBS1h28077f2dbe1c` (`e3/sessions/`)
outcome	416 records. Full-solve arms: 248/248 verified (`[d]G = Q`). Capped F4 arms: 168/168 `exhausted` by design
aggregation	`examples/generator_effect_e3_summary.rs` → `e3/summary.json`; pricing through `ecbench::band::{unpriced_total, calibrated_native}`; 10 000 resamples, seed 20261019
extra inputs	`e3/sizes.jsonl` (B, N_3; `generator_effect_e3`), `e3/f4cost.jsonl` (146 F4 systems, planted and random; `generator_effect_e3 f4cost`)
graphs	`e3/e3.svg`, `e3/e3_f4.svg` (PNG beside them), from `e3/plot_e3.py`, which plots `summary.json` only
run report	`reports/run-20261019T2334Z/`

Method: what is measured and what is substituted

- **Measured:** every full-solve arm (`sem-grid`, `kl-mitm3`, `rho-neg`). F4′s multiply-adds per call are measured on the capped runs: 3,200 / 800 / 80 / 8 calls per curve at t = 2 / 3 / 4 / 5. The planted-system spread is measured by `f4cost`.
- **Substituted (not run):** the F4 arms' full-solve cost. A full F4 solve needs about `(B + 1)/h` trials, about 10^5 at t = 4 at about 1 s each, which is out of reach. `sem-chain`, `sem-full` and `sem-grid` share base and seed, so they draw the same trial stream and reach the same verdicts. So each grid record is re-priced per target: the grid multiplications on refuted trials are replaced by F4′s mean multiply-adds per call plus the system-specialisation multiplications per call. The grid work on consistent trials stays in, because the F4 arm reads solutions off the grid too. Group work is the grid record's. The two facts the substitution rests on were checked:
  - **Same stream:** the chain and full capped records have identical group work and identical trial counts on every curve and t. All 168 capped records report 0 F4/grid disagreements over 24,502 F4 calls, all refutations and every one grid-checked (`pkm_crosscheck_disagreements = 0`).
  - **Cost by shape:** every F4 refutation at a given (t, presentation) costs the same multiply-adds. The count is identical across targets and across the three curves:

t	chain	full
2	$3.908 \cdot 10^5$	$2.243 \cdot 10^5$
3	$1.587 \cdot 10^8$	$1.260 \cdot 10^7$
4	$1.060 \cdot 10^{10}$	$1.683 \cdot 10^{10}$
5	$3.186 \cdot 10^{12}$	—

The capped mean equals the `f4cost` refutation count to 4 digits. Consistent (planted) systems cost 0.53–1.52× a refutation, at most 1.52× (P2: ≤ 2×). Solving degree is 5 / 6 / 6 / 7 for chain and 7 for full.

- **Pricing:** `S(w) = (G − C + w·(U + n_C)) / √r` with `U` every unpriced counter of the record. For F4 that is the solver block's multiply-adds; for the grid, `pkm_grid_fp_mults_uncharged`; for MITM, lookups and target-guard probes; for the base build, square roots and Legendre symbols. `C = n_C = 0`: these fresh curves have no pinned calibration.

Results

1. Yield at equal B (P1 holds)

N_3 (exact) is identical for the two bases in all 12 cells, from $N_3 = 4$ (e3-c3, t = 2, B = 1) to $N_3 = 9196$ (t = 5, B = 19). The measured hit rates (Wilson 90%) of `sem-grid` and `kl-mitm3` cover N_3 / r in all 24 (curve, t, arm) cells. Examples: e3-c1 t = 5, h = $1.015\cdot10^{-3}$; grid $1.110\cdot10^{-3}$ [0.971, 1.27]$\cdot10^{-3}$; known-log $1.505\cdot10^{-3}$ [0.848, 2.67]$\cdot10^{-3}$. The one miss is on the 12-bit control curve, outside the population: `kl-mitm3` found 8 relations in 170 trials against `h = 0.0185`, and its interval [0.027, 0.082] misses `h` (see "Controls").

2. S at the band weights, and the flip (P3 holds)

S ratio of the Semaev arm (F4 chain, substituted) to `kl-mitm3`, paired over 8 targets, with 90% intervals:

curve	t	B	w = 0	w = 0.1	w = 1	w* (F4 chain)
e3-c1	2	2	0.35 [0.18, 0.62]	$1.13\cdot10^4$ [0.61, 1.99]$\cdot10^4$	$6.2\cdot10^4$	$5.2\cdot10^{-6}$
e3-c1	3	4	0.52 [0.36, 0.78]	$7.4\cdot10^6$ [5.1, 11]$\cdot10^6$	$4.1\cdot10^7$	$5.9\cdot10^{-9}$
e3-c1	4	9	0.65 [0.36, 1.32]	$6.1\cdot10^8$ [3.4, 12]$\cdot10^8$	$3.4\cdot10^9$	$5.2\cdot10^{-11}$
e3-c1	5	19	0.68 [0.42, 1.38]	$1.8\cdot10^{11}$ [1.1, 3.5]$\cdot10^{11}$	$1.0\cdot10^{12}$	$1.6\cdot10^{-13}$
e3-c2	2	3	0.68 [0.42, 1.17]	$2.3\cdot10^4$ [1.4, 3.9]$\cdot10^4$	$1.3\cdot10^5$	$1.3\cdot10^{-6}$
e3-c2	3	5	0.55 [0.37, 0.79]	$7.8\cdot10^6$ [5.2, 11]$\cdot10^6$	$4.3\cdot10^7$	$5.3\cdot10^{-9}$
e3-c2	4	10	0.47 [0.29, 1.01]	$4.5\cdot10^8$ [2.7, 9.5]$\cdot10^8$	$2.5\cdot10^9$	$1.1\cdot10^{-10}$
e3-c2	5	19	0.44 [0.24, 1.19]	$1.2\cdot10^{11}$ [0.63, 3.1]$\cdot10^{11}$	$6.4\cdot10^{11}$	$4.5\cdot10^{-13}$
e3-c3	2	1	0.72 [0.43, 1.34]	$1.9\cdot10^4$ [1.2, 3.4]$\cdot10^4$	$1.1\cdot10^5$	$1.3\cdot10^{-6}$
e3-c3	3	3	0.48 [0.25, 1.21]	$6.6\cdot10^6$ [3.4, 17]$\cdot10^6$	$3.7\cdot10^7$	$7.1\cdot10^{-9}$
e3-c3	4	9	0.63 [0.45, 0.96]	$5.9\cdot10^8$ [4.2, 9.0]$\cdot10^8$	$3.3\cdot10^9$	$5.7\cdot10^{-11}$
e3-c3	5	18	0.30 [0.19, 0.60]	$8.0\cdot10^{10}$ [5.0, 16]$\cdot10^{10}$	$4.5\cdot10^{11}$	$7.9\cdot10^{-13}$

- **At w = 0** F4 is free, and the Semaev arm pays only for target draws and lifts. It collects `B + 1` relations at about one addition per trial, where known-log MITM pays 2B additions per trial for one relation. Every point estimate is below 1. The interval excludes 1 in 5 of 12 cells and contains it in 7. `sem-full` gives the same w = 0 ratios, because its group work is the same.
- **At w = 0.1 and w = 1** every cell is above $6\cdot10^3$. At t ≥ 3 the chain cells are above $6\cdot10^6$ and the full cells above $5\cdot10^5$: P3′s 10^4 bar with room to spare. The full presentation is 12.6× cheaper than chain at t = 3 and 1.6× dearer at t = 4. Neither changes the picture.
- **§5a verdict:** the order of the Semaev arm and known-log MITM changes inside the band in every cell. **Unresolved. Not a win for either.**

3. Crossover in B (P4 holds)

- **At w = 0.1 and w = 1:** the Semaev arm is above known-log at every t on every curve, so no `B*` exists.
- **At w = 0:** it is below at every t, so again there is no crossover.
- **`w*`:** it falls monotonically along t on every curve, by 49× to 880× per level, and is below 10^{-8} for every t ≥ 3.

- **Why:** F4′s multiply-adds per call grow 406× (t = 2⟶3), 67× (3⟶4) and 301× (4⟶5), while MITM's per-trial cost doubles. The yield is identical at equal B, so the gap between the arms is the per-trial decision cost alone. At w = 0.1 it grows from about 10^4 to 10^{11} across the band.
- **Extrapolation:** `t = 6` (N = 18) was not run, and the PKM note §16 put one system there at 44 minutes and its host's 14 GB. Extending the measured per-level ratios gives no sign of a turn. This is an extrapolation.

4. The grid reference (P5 holds)

The grid decides the same systems with 244 / 1,224 / 7,440 / 51,232 `F_p` multiplications per call at t = 2…5, identical on every curve. F4 takes $1.6{\cdot}10^3$, $1.3{\cdot}10^5$ (full: $1.0{\cdot}10^4$), $1.4{\cdot}10^6$ and $6.2{\cdot}10^7$ times more. The grid arm is itself measured end to end:

- **At w = 0:** below known-log, with S ratio 0.30–0.72.
- **At w > 0:** above it, from 7.4× at t = 2 (e3-c1, w = 0.1) to $1.6{\cdot}10^4$× at t = 5 (w = 1).

Its `O(|V|³)` cost per target grows faster than MITM's `O(B)`.

5. Distance to rho (P6 holds; reported as distance, not a gate)

Every IC arm is above `rho-neg` in every cell and at every weight. The closest is at t = 5:

- **grid at w = 0:** 5.5–7.0× rho;
- **known-log MITM:** 8.6–23× rho at every weight.

F4 at w = 0.1 is 10^{11}–10^{12} times above rho at t = 5. Rho's S is 1.08 / 1.18 / 2.04 (e3-c1), 1.33 / 1.45 / 2.56 (e3-c2) and 1.17 / 1.28 / 2.24 (e3-c3) at w = 0 / 0.1 / 1.

Controls

- **End to end, F4 actually run:** on the 12-bit control curve, `sem-chain-t2` ran F4 on every trial to full solves: 8/8 verified, 1,228 F4 calls, 0 disagreements. Its S equals `sem-grid-t2`′s at w = 0 (18.38), as the substitution assumes. Its measured F4 count, $3.927{\cdot}10^5$ per call, matches the 24-bit refutation count to 0.5%. Its `ecbench band` price, measured, is 18.383 / 123,668 / 1,236,514 at w = 0 / 0.1 / 1. The substitution formula applied to the control's grid records gives the same three numbers. This checks the substitution's accounting. It does not check the cost-by-shape step, because there the mean comes from the same calls.
- **The control's `kl-mitm3` hit rate:** 8 hits in 170 trials against `h = 0.0185`; the Wilson interval misses `h`. The stream stops at the first relation, so the 8 samples are geometric waiting times and a fixed-n interval is only approximate for them. The walk also revisits points on a group of 2,377 elements, and the target guard skips the repeats. It is reported as measured. On this curve the Semaev arm is above known-log even at w = 0 (1.28 [1.06, 1.55]).
- **Unit and harness tests (this PR):**
  - planted `m = 3` decompositions found in both presentations and by the grid;
  - F4 against the grid and the exact decomposable set on random and planted targets;
  - refusal of a foreign base;
  - a verified `solve_ic_prime` run whose query stream and relations match the grid.

Predictions against outcome

	prediction	outcome
P1	N_3 equal within 1%; hit rates cover N_3/r	holds: identical N_3 in 12/12 cells; 24/24 intervals cover
P2	refutations identical per shape; consistent ≤ 2×	holds: identical to the last digit across targets and curves; consistent ≤ 1.52×
P3	below at w = 0, above ≥ 10^4 at w ≥ 0.1 and t ≥ 3; flips, unresolved	holds in point estimates; at w = 0 the interval excludes 1 in 5/12 cells
P4	w^* falls with B, < 10^{-4} for t ≥ 3, no B^*	holds: w^* < 10^{-8} for t ≥ 3, no B^* at any weight
P5	grid ≥ 10^3 cheaper than F4 per trial for t ≥ 3	holds: $1.0{\cdot}10^4$ to $6.2{\cdot}10^7$ ($1.6{\cdot}10^3$ at t = 2)
P6	every IC arm above rho	holds

Scope and limits

- **Population:** a restricted prime family (`2^t | p − 1`), three curves, `m = 3`, `t ≤ 5`, `B ≤ 19`. These B are far below E1′s MITM optimum (`B ≈ r^{1/3} ≈ 210`), because F4 does not reach larger towers on this host.
- **Engine:** one engine (`f4_fp_tower`, normal strategy, degree bound 12) and two presentations. A different engine or a different system (for example symmetrised variables, or an isogeny tower) could change F4′s constants; the measured growth per level makes a turn inside this band unlikely.

- **Substitution:** the F4 arms' full-solve S is a substitution on measured records, not a run. It is checked by identical streams and verdicts, by shape-determined per-call counts, and by the control curve, where F4 ran to full solves.
- **No security statement.** These are generic precomputation shapes and no exponent moves. This is not an improvement over the IC baseline: the Semaev arm is above known-log at every $w > 0$, and unresolved across the band.

Files

- `src/cryptanalysis/ic_framework/pkm_semaev.rs`: the oracle, the grid reference and `decomposable_points`, with unit tests.
- `src/cryptanalysis/ecbench/methods.rs`: registration and the end-to-end test.
- `src/cryptanalysis/ecbench/claim.rs`: claim entry.
- `examples/generator_effect_e3.rs`: sizes, N_3 and `f4cost`.
- `examples/generator_effect_e3_summary.rs`: aggregation.
- `examples/a_minus_3_prime_order_curves.rs`: gains the optional `MIN_V2` filter.
- `e3/specs/`, `e3/sessions/`, `e3/summary.json`, `e3/plot_e3.py`, figures.
- `E3.pdf` (this report), built by `build_e3_pdf.sh`.